



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

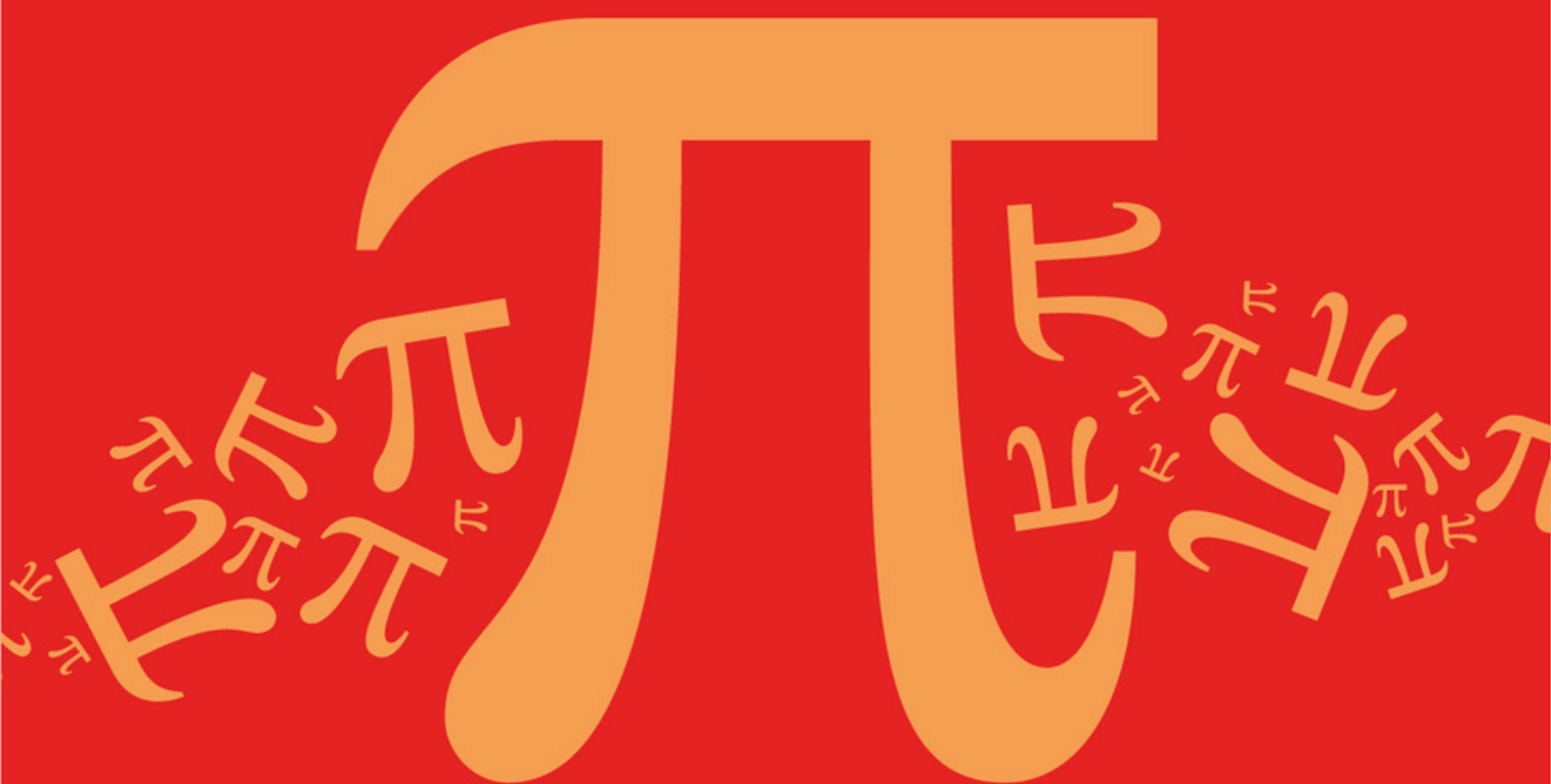
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DA: Graphs

Sub topic

DA5: Complete, Bipartite, and Complement Graphs

Mark Scheme

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Mark Scheme

Topic

DA: Graphs

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DA5: Complete, Bipartite, and Complement Graphs

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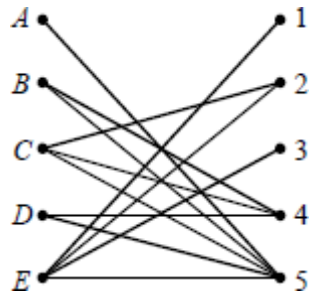
**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a)



Bipartite graph, 2 sets of 5 vertices, at least 9 edges

All correct, including labelling

M1

A1

2

(b) Only *E* can do task 1 and task 3.

M1

One person cannot do 2 tasks so impossible.

A1

2

Or

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A does 5, then

B must do task 4 and *D* must do task 4.

*Must have A to 5 first, or 3 people A, B, D
can only do 2 tasks 4, 5*

(M1)

One task cannot be done by 2 people so impossible.

Not enough tasks for the number of people so impossible.

(A1)

Or

4 people *A, B, C, D* can only do 3 tasks 2, 4, 5

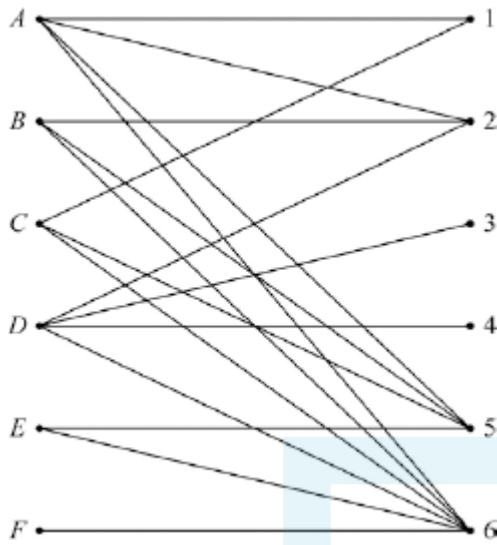
(M1)

Not enough tasks for the number of people so impossible.

(A1)

[4]

Q2.



Bipartite graph, 2 sets of 6 vertices, at least 10 edges

Correct, including labels

M1

A1

[2]

Q3.

	1	2	3	4	5	6
A	0	0	1	0	0	0
B	1	0	0	1	0	0
C	0	0	1	0	0	1
D	1	1	0	0	0	0
E	0	0	0	1	1	0
F	0	0	0	0	1	1

*6 × 6 matrix labelled with some 0, 1, ✓, ✗'s
(at least 9 entries)*

All correct

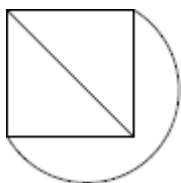
M1

A1

[2]

Q4.

(a)



E1

1

(b) (i) 28

B1

(ii) Odd number of edges at (all) vertices
Must see the word odd, not just 7

E1

2

(c) (i) $\frac{n(n-1)}{2}$ OE

B1

(ii) $n - 1$

B1

(iii) n must be odd
Must have n in their answer

E1

(iv) $n = 3$ ©2025 Exam Papers Practice. All rights reserved.

Must have n in their answer

B1

4

[7]

Q5.

	1	2	3	4	5	6
A	0	0	0	1	1	0
B	0	0	1	0	1	1
C	0	0	0	1	0	0
D	0	1	0	0	0	1
E	0	1	0	1	0	0
F	1	0	1	0	1	0

(6 x 6) matrix labelled with some $\surd$'s or $\times$'s or 0's
 or 1's or – 's

M1

CAO

A1

[2]

Q6.

(a) (i) 10

B1

1

(ii) 4

B1

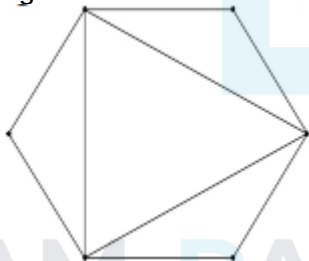
1

(iii) 5

B1

1

(b) eg.



Simple graph, 6 vertices

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Eulerian graph with 9 edges

M1

A1

2

[5]

Q7.

(a)



Correct graph

4 vertices

M1

A1

2

(b) (i) 15

B1 1

(ii) 5

B1 1

(iii) No, order of vertices is odd

B1 1

(c) (i) $\frac{n(n-1)}{2}$
OE

B1 1

(ii) $n - 1$

B1 1

(iii) n ODD
OE

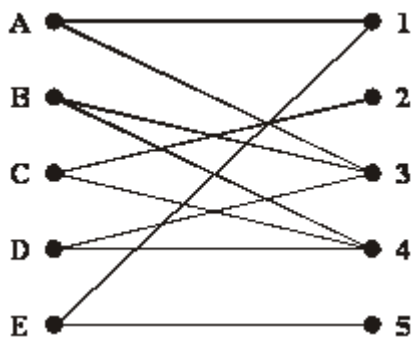
B1 1

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[8]

Q8.



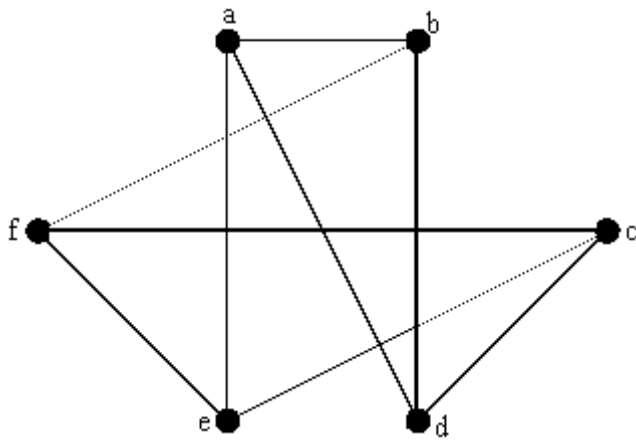
At least 8 correct links; not more than 1 incorrect link

M1 A1

[2]

Q9.

(a)

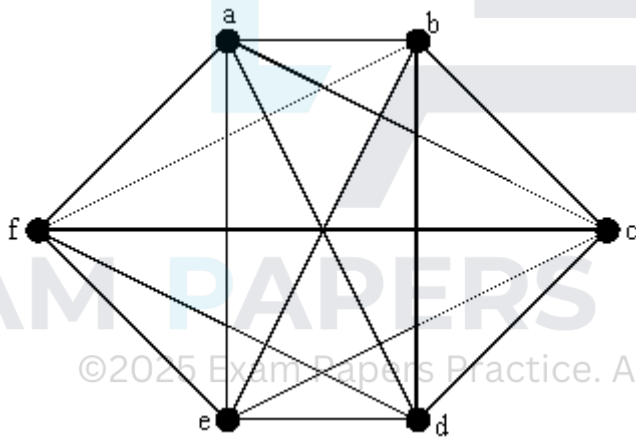


M1

A2 –1 each error

3

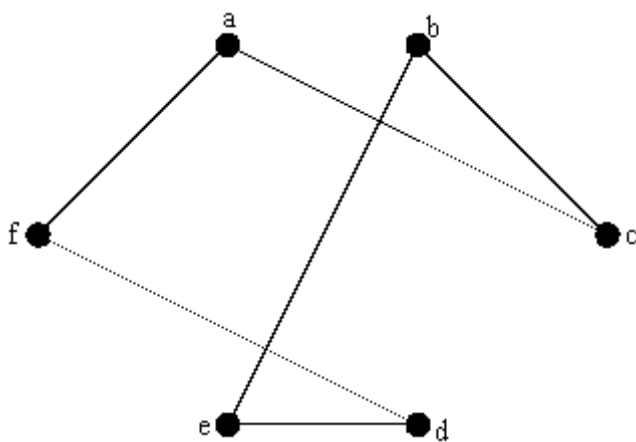
(b)



B1

1

(c)



M1

A1

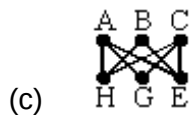
Indicates available matchings.

B1

3

[7]

Q10.



$K_{3,3}$

correct

B1

B1

M1

A1

[4]

Examiner reports

Q1.

In part (a), apart from the rare slip or completely wrong format this was almost always correctly answered. Most students knew the answer to part (b), but many were unable to express it accurately. Many either stated the relevant specific links without explaining why they made a complete matching impossible, or made general statements about what constitutes a complete matching without tying it to the given situation.

Q2.

Almost all candidates scored full marks on this question – the common error was to omit the link from D to 6.

Q3.

The majority of students scored well on this question.

Q4.

This question differentiated well between the students. Many students showed a lack of understanding or knowledge of graph theory and were unable to answer this question correctly and weaker students found the abstractness of this question a too much of a challenge.

The most common problem with par (b)(ii) was an incomplete or inaccurate description of vertices of an odd order, with students often talking about “odd vertices” or an “odd number of vertices”.

Marks were quite often lost in part (c) because of a failure to note the implications of the phrase “state in terms of n ” and part (c)(iv) identified the strongest students.

Q5.

Apart from the rare slip or completely wrong format, this question was almost always correct.

Q6.

Very few candidates failed to score marks but very few scored full marks. The first mark was often dropped in part (a). Few candidates got both marks in part (b) - many joined two vertices with more than one edge; a majority of those who surmounted this hurdle then presented vertices of odd order.

Q7.

This question was the least well answered on the paper. Although candidates must be able to apply their knowledge in questions, this tested graph theory itself and not the application of graph theory. The majority of candidates were able to answer parts (a), (b)(ii) and (c)(ii). Complete networks were not understood. The total number of edges was poorly answered. Parts (b)(iii) and (c)(iii) tested the understanding of Eulerian graphs and whilst candidates realised that the order of vertices was important, many failed to score the mark in part (c)(iii).

Q8.

This was usually done well, but many candidates failed to score full marks as their alternating path was not clearly shown. It is strongly recommended that if candidates work on their bipartite graph, they also list the alternating path(s).

Q9.

This question caused candidates few problems.

Q10.

Almost all candidates could correctly draw the graphs K_5 and $K_{3,3}$. Most reasoned that they were looking for a subgraph isomorphic to $K_{3,3}$ rather than K , in part (c), and were credited for that reasoning even if they could not find the required subgraph.



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